



Bahir Dar University
Bahir Dar University Institute of Technology
Faculty of Chemical and Food Engineering
Chemical Engineering Department Holistic
Exam for 2023

- ❖ Target Group - 4th Year Chemical Engineering Students
- ❖ Time Allowed - Three (3:00) Hours

Student Name _____ **ID No** _____.

Instructions;

- ✚ Turn off your mobile phone.
- ✚ Open your exam paper when you are told to do so.
- ✚ Make sure you write your name on the exam and answer key paper.
- ✚ Once the examination has begun, leaving the examination room before one hour is not allowed.
- ✚ If you have time (when you finish the exam before time) check your answers.
- ✚ Please read all questions carefully and choose the best answer & write the final answer In **Capital Letter** on the answer sheet paper (last page).

Good luck!

1. Which of the following represents heterogeneous catalytic reaction?
 - A. Burning of coal
 - B. Roasting of ores
 - C. Ammonia synthesis
 - D. Reduction of iron ore
2. The rate of a chemical reaction tells us about
 - A. The reactants taking part in the reaction
 - B. The products formed in the reaction
 - C. How slow or fast the reaction is taking place
 - D. All
3. When the rate of the reaction is equal to the rate constant, the order of the reaction is
 - A. Zero order
 - B. First order
 - C. Second order
 - D. Third order
4. A catalyst alters, which of the following in a chemical reaction?
 - A. Entropy
 - B. Enthalpy
 - C. Internal energy
 - D. Activation energy
5. Which of the following theories does not propose the temperature dependence of rate constant?
 - A. Arrhenius theory
 - B. Collision theory
 - C. Transition state theory
 - D. Bohr's theory
6. Which of the following is NOT a part of reactor design?
 - A. Size of the reactor
 - B. Type of the reactor
 - C. Flow rates of the reactant and product streams
 - D. Mode of operation
7. According to tanks in series model, the spread of the tracer curve is proportional to
 - A. Square of distance from the tracer origin
 - B. Square root of distance from the tracer origin
 - C. Cube of distance from the tracer origin

- D. Inverse Square of distance from the tracer origin.
8. Selectivity in parallel reactions is defined as the ratio of
 - A. Moles of desired product formed to the moles of undesired material formed
 - B. Moles of undesired product formed to the moles of desired material formed
 - C. Moles of product formed to the moles of reactant consumed
 - D. Moles of reactant consumed to the moles of product formed
 9. Which of the following is not the type of energy?
 - A. Entropy
 - B. Heat
 - C. Enthalpy
 - D. D. Work
 10. What are the degrees of freedom?
 - A. Total variables in the process
 - B. Total species in the process
 - C. Total reactions in the process
 - D. None of the mentioned
 11. Which of the following is not an example of mechanical energy?
 - A. Work
 - B. Kinetic energy
 - C. Potential energy
 - D. Internal energy
 12. Which of the following is constant in a closed system?
 - A. Energy
 - B. Mass
 - C. Temperature
 - D. Momentum
 13. How long would it take for a reversible process to complete?
 - A. Instant
 - B. Infinite
 - C. Depends on the process
 - D. None of the mentioned
 14. What is ΔU for a steady-state system?
 - A. Positive
 - B. Negative
 - C. Zero
 - D. Cannot say
 15. Total energy of the universe is always
 - A. Increasing

- B. Decreasing
C. Constant
D. Cannot say
16. Which of the following does the concept of material balance based upon?
- A. Conservation of mass
B. Conservation of energy
C. Conservation of momentum
D. Conservation of Volume
17. Which one of the following is process by which particles are joined or bind with one another in a random way without using agitation and ending with an aggregate of porous structure?
- A. Agglomeration
B. Granulation
C. Wetting
D. Mixing
18. Which of the following device used for mixing, granulation and drying purpose?
- A. Mixer Granulator
B. Tumbling Granulator
C. Fluidized bed granulator
D. All
19. If a steel ball with 3mm diameter is allowed to settle freely in water which starts initially from rest, what will be its terminal velocity? (Take the density of the ball 1.75 g/cm^3 , density of fluid 1 g/cm^3 , viscosity of fluid 0.002 Ns/m^2 and gravity of earth 9.81 m/s^2 & Reynolds number respect to the particle is 800)
- A. 0.54 m/s
B. 0.82 m/s
C. 0.26 m/s
D. 0.94 m/s
20. Which of the following is not categorized as a "mechanical operation"?
- A. Humidification
B. Agitation
C. Filtration
D. Size enlargement
21. For the following particulate material distribution what will be the mass mean diameter?

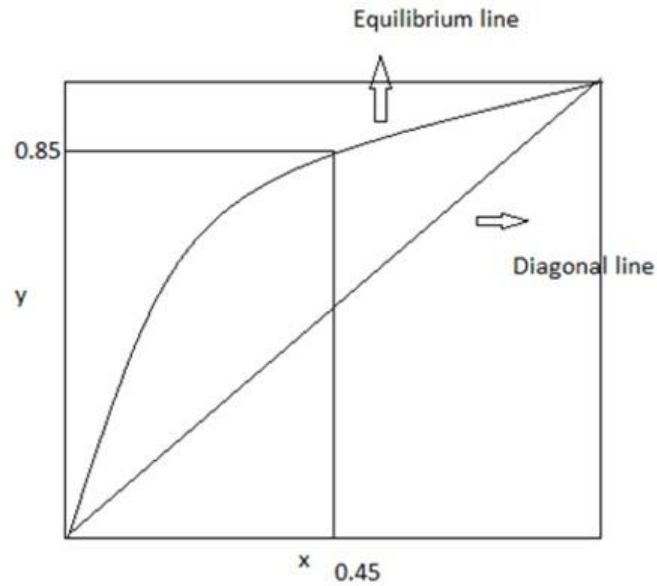
Size range in μm	-704+352	-352+176	-176+88	-88+44	Pan
Mass in gm	25	37.5	62.5	75	50

- A. 170.20 μm
B. 120.60 μm
C. 134.50 μm
D. 149.10 μm

22. By which approach can we estimate the pressure drop per unit length of the bed during packed bed column, if the Reynolds number of the flow ($Re < 10$)?
- A. Carman-Kozeny approach
 - B. Burke Plummer approach
 - C. Ergun approach
 - D. All
23. The specific surface of spherical particles is proportional with D_p as; (where, D_p is diameter of particle)
- A. $1/D_p$
 - B. D_p^2
 - C. $C. D_p$
 - D. $1/D_p^2$
24. Which one of the following device is used for the transportation of ultrafine particles with a complete enclosure?
- A. Screw conveyor
 - B. Belt conveyor
 - C. Chain conveyor
 - D. Pneumatic conveyor
25. Which one of the following unit operation involves manipulation of a heterogeneous physical system with the intent to make it more homogeneous?
- A. Agglomeration
 - B. Granulation
 - C. Filtration
 - D. Mixing
26. Heat is transferred by all three modes of transfer in
- A. Electric Heater
 - B. Steam Condenser
 - C. Melting of Ice
 - D. Boiler
27. Two balls of same material and finish have their diameters in the ratio of 2:1 and both are heated to same temperature and allowed to cool by radiation. Rate of cooling by big ball as compared to smaller one will be in the ratio of
- A. 2: 1 B. 1: 2 C. 4: 1 D. 1: 4
28. Fourier's law of heat conduction is valid for

- A. One dimensional cases only
 - B. Two dimensional cases only
 - C. Three dimensional cases only
 - D. Regular surfaces having non-uniform temperature gradients
29. Methane diffuses at steady state through a tube containing helium. At point 1 the partial pressure of methane is $p = 55 \text{ kPa}$ and at point 2, 0.03 m apart $P = 15 \text{ KPa}$. The total pressure is 101.32 kPa , and the temperature is 298 K . At this pressure and temperature, the value of diffusivity is $6.75 \times 10^{-5} \text{ m}^2 / \text{sec}$. what will be the flux of CH_4 at steady state for equimolar counter diffusion?
- A. $3.666 \times 10 \text{ kmol/ m}^2 \text{ sec}$
 - B. $3.866 \times 10 \text{ kmol/ m}^2 \text{ sec}$
 - C. $3.833 \times 10 \text{ kmol/ m}^2 \text{ sec}$
 - D. $3.633 \times 10 \text{ kmol/ m}^2 \text{ sec}$
30. Heat transfer in a long, hollow cylinder which is maintained at uniform but different temperatures on its inner and outer surfaces may be assumed to be taking place in which direction?
- A. Axial only
 - B. Unpredictable
 - C. Radial only
 - D. No heat transfer takes place
31. The appropriate rate equation for convective heat transfer between a surface and adjacent fluid is prescribed by
- A. Newton's first law
 - B. Wein's displacement law
 - C. Kirchhoff's law
 - D. Newton's law of cooling
32. Regarding one dimensional heat transfer, choose the correct statement.
- A. Steady – $f(x)$, Unsteady – $f(x, t)$

- B. Steady – $f(x, t)$, Unsteady – $f(x)$
 - C. Steady – $f(x, y, t)$, Unsteady – $f(x)$
 - D. Steady – $f(y, z)$, Unsteady – $f(y)$
33. Which of the following would not lead to a reduction in thermal resistance?
- A. In conduction, reduction in the thickness of the material and an increase in thermal conductivity
 - B. In radiation, increasing the temperature and the emissivity
 - C. In radiation, increasing the temperature and reducing the emissivity
 - D. In convection, stirring of the fluid and cleaning the heating surface.
34. Which one of the following statement is correct?
- A. Diffusivity decreases with increase in temperature
 - B. Diffusivity increases with increase in molecular weight
 - C. Diffusivity increases with the size of the individual molecule
 - D. Diffusivity increases with decrease in viscosity of liquid
35. Mass transfer rate between two fluid phases does not necessarily depend on the _____ of the two phases.
- A. Chemical properties
 - B. Physical properties
 - C. Degree of turbulence
 - D. Interfacial area
36. In McCabe-Thiele method, at infinite reflux ratio
- A. number of plates is maximum
 - B. the operating lines coincide with the diagonal
 - C. bottom product is maximum
 - D. overhead product is maximum
37. Find the relative volatility, if x and y are the concentrations in mole fraction



- A. 6.92 B. 0.14 C. 1.88 D. 2.88

38. Which of the following processes will be occurred when unsaturated air (with dry bulb temperature 12°C and relative humidity 47%) passing through water spray chamber (temperature of water is 40°C)?
- A. Humidification only
 - B. Heating only
 - C. Both heating and humidification
 - D. Evaporative cooling
39. Which of the following dryer is used to produce powder from the solution?
- A. Spray dryer
 - B. Cabinet tray dryer
 - C. Pneumatic dryer
 - D. Fluidized bed dryer
40. Solvent extraction is the terminology applied to the liquid-liquid extraction, which is preferred for the separation of the components of liquids, when
- A. Extracting solvent is cheaply and abundantly available
 - B. One of the liquid components is heat sensitive

- C. Viscosity of liquid components is very high
- D. One of the liquid components has very high affinity towards the solvent

41. What happens in an ideal leaching stage of a mixture of A and B?

- A. The solvent dissolves Some B and some A
- B. The solvent dissolves none of A and all B
- C. The solvent dissolves all of A and B, which makes separation easier
- D. The solvent does not dissolve any of A and B

42. The specific heat capacity at constant pressure is greater than the specific heat capacity at constant volume for an ideal gas with exception of water at 4°C. This may due to

- A. All the heat supplied at constant pressure raises its internal energy and does work in expanding the gas against the external pressure
- B. All the heat supplied at constant pressure only raises its internal energy
- C. All the heat supplied at constant volume raises its internal energy and does work in expanding the gas against the external pressure
- D. All the heat supplied at constant pressure only does work in expanding the gas against the external pressure

43. Which statements best describes thermodynamics?

- A. Introduces the concept of temperature and thermodynamic equilibrium where two objects attain the same temperature when brought in thermal contact.
- B. Studies the relationships involving heat, mechanical work and other aspects of energy transfer that takes place in thermodynamic systems.
- C. Illustrate no cyclic process can transfer heat from a cold reservoir to a hotter region with no input of mechanical work.
- D. States that the efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.

44. Rate of energy increase within the control volume is given by

- A. Rate of energy inflow * rate of energy outflow
- B. Rate of energy inflow – rate of energy outflow
- C. Rate of energy inflow / rate of energy outflow

- D. Rate of energy inflow + rate of energy outflow
45. In an ideal solution, the activity of a component equals its
- A. Mole fraction
 - B. Fugacity at the same temperature and pressure
 - C. Partial pressure
 - D. Fugacity at same pressure only
46. A househusband is cooking beef stew for his family in a pan that is (I) uncovered, (II) covered with a light lid, and (III) covered with a heavy lid. For which case will the cooking time be the shortest?
- A. I
 - B. II
 - C. III
 - D. I & III
47. A Carnot engine, operating between reservoirs at 20°C and 200°C, produces 10 kW of power. The rejected heat is nearest:
- A. 26.3 kJ/s C. 16.3 kJ/s
 - B. 20.2 kJ/s D. 12.0 kJ/s
48. If heat Q flows reversibly from the system to the surroundings at ' T_o ', what will be the effect on entropy?
- A. Entropy of the system is reduced by Q/T_o
 - B. System has lost entropy to the surroundings
 - C. Entropy increase of the surroundings is Q/T_o
 - D. All of the mentioned
49. Suppose you have a Double Pipe Heat Exchanger, with the inner tube of diameter 20mm (neglect thickness) and outer tube of diameter 30mm. You have also two fluids A & B, you desire to have their flow rates as 15 m³/s and 21 m³/s respectively. Which fluid pass arrangement would be best suited for the heat exchanger?
- A. Parallel flow with A in inner pipe and B in the annulus

- B. Counter flow with B in inner pipe and A in the annulus
 - C. Counter flow with A in inner pipe and B in the annulus
 - D. Parallel flow with B in inner pipe and A in the annulus
50. A decision must be made as to which fluid will flow through the tubes and which will flow through the shell. As a professional engineer, which fluid allocation you recommended
- A. The more seriously fouling fluid flows through the tube
 - B. The corrosive fluid must flow on the shell side
 - C. The stream with the lower heat transfer coefficient flows through the tube
 - D. The more seriously fouling fluid flows on shell side.
51. When gas is used as a fluid in a double pipe heat exchanger, which one of the following is not true?
- A. Fins increase necessary heat transfer area
 - B. The gas side heat transfer coefficient is the highest
 - C. Extended fins are used on the gas side to increase the Heat Transfer coefficient
 - D. The gas side has a low heat transfer coefficient
52. When a fluid is used in a Shell and Tube heat exchanger, which one of the following is not true?
- A. Fins increase necessary heat transfer area
 - B. If the fluid is gas then the gas side heat transfer coefficient is the lowest
 - C. Extended fins are used on the shell side to increase the Heat Transfer coefficient
 - D. Baffles are provided only to work as fins\
53. Avoiding the Critical heat flux is an engineering problem in heat transfer applications, such as nuclear reactors, where fuel plates must not be allowed to overheat. To cope with this, what measure can be taken?
- A. Add more water
 - B. Add turbulence
 - C. Increase the pressure
 - D. Decrease the pressure

54. Which of the following phase of fluid need much higher heating surface area.
- A. Gas–Liquid phase
 - B. Liquid –Liquid phase
 - C. Gas–Gas phase
 - D. Solid- liquid phase
55. Which one of the following do we prefer to use for single effect over multiple effect evaporator?
- A. Low pressure operations
 - B. High pressure operations
 - C. Corrosive materials use
 - D. High temperature operations
56. Forced circulation evaporators are usually more costly than natural circulation evaporators. However, the natural circulation evaporators are not suitable **except**.
- A. Highly viscous solutions due to low heat transfer coefficient
 - B. Solution containing suspended particles
 - C. For heat sensitive materials
 - D. Low viscous solutions due to high heat transfer coefficient
57. Newton’s law of viscosity relates
- A. intensity of pressure and rate of angular deformation
 - B. shear stress and rate of angular deformation
 - C. shear stress, viscosity and temperature
 - D. viscosity and rate of angular deformation

58. In a forced vortex motion, the velocity of flow is
- A. Directly proportional to its radial distance from axis of rotation
 - B. Inversely proportional to its radial distance from the axis of rotation
 - C. Inversely proportional to the square of its radial distance from the axis of rotation
 - D. Directly proportional to the square of its radial distance from the axis of rotation
59. The shear stress distribution for a fluid flowing in between the parallel plates, both at rest, is
- A. Constant over the cross section
 - B. Parabolic distribution across the section
 - C. Zero at the mid plane and varies linearly with distance from mid plane
 - D. Zero at plates and increases linearly to midpoint
60. The velocity distribution for laminar flow through a circular tube
- A. Constant over the cross-section
 - B. Varies linearly from zero at walls to maximum at centre
 - C. Varies parabolically with maximum at the centre
 - D. Varies parabolically with minimum at the centre
61. In series-pipe problems
- A. The head loss is same through each pipe
 - B. The discharge is same through each pipe
 - C. A trial solution is not necessary
 - D. The discharge through each pipe is added to obtain total discharge
62. The major loss of energy in long pipes is due to
- A. Sudden enlargement
 - B. Sudden contraction
 - C. Gradual contraction or enlargement
 - D. Friction